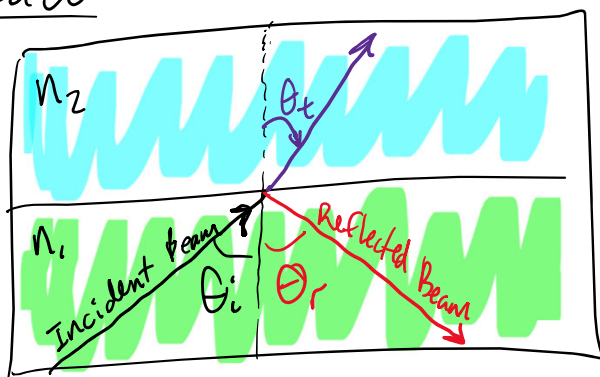


Snell's Law



*Note θ is always with respect to the vertical

* B/c n is directly tied to the velocity of the EM wave in the material

$$\frac{v_1}{v_2} = \frac{\left(\frac{1}{\sqrt{\epsilon_{r1}\epsilon_0\mu_0}}\right)}{\left(\frac{1}{\sqrt{\epsilon_{r2}\epsilon_0\mu_0}}\right)} = \frac{\left(\frac{c}{\sqrt{\epsilon_{r1}}}\right)}{\left(\frac{c}{\sqrt{\epsilon_{r2}}}\right)} = \frac{\sqrt{\epsilon_{r2}}}{\sqrt{\epsilon_{r1}}} = \frac{n_2}{n_1} = \frac{\sin \theta_i}{\sin \theta_t}$$

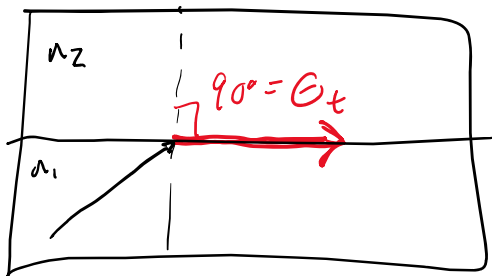
$$(n_i)(\sin \theta_i) = (n_t)(\sin \theta_t)$$

$$n_2 = n_t$$

$$n_1 = n_i$$

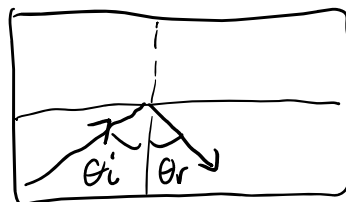
* When θ_t reaches "90°" the incident angle (θ_i) becomes "critical angle" θ_c

$$\sin \theta_c = \frac{n_t}{n_i}$$



for all incident beams $\theta_i > \theta_c$ there is no transmission into (n_t) $\rightarrow$ only reflection.

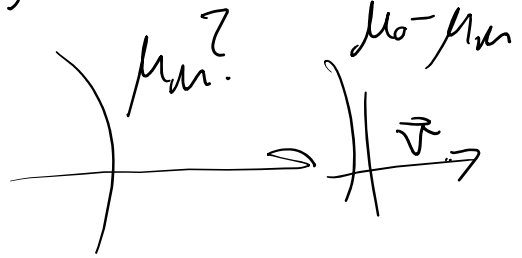
Total Internal Reflection (T.I.R)



T.I.R * very important for

- fiber optic cables
- liquid spectroscopy.

~~μ_{metal}~~ $\rightarrow$ does not exist



T.I.R.



*No transmitted wave

*at a distance d (Not touching)
a transmitted wave (beam)
appears!

* why does it happen?

